

SARAD PATEL UNIVERSITY

B.C.A. – 1st Semester (CBCS) NC

Examination - February, 2020

US01EBCA01: Digital Computer Electronics

Date: 11/02/2020

Time: 10.00 Am to 12.00 pm

Marks: 70

Q1. Select the correct option for the following questions.

[10]

1. An invert gate is also called a _____ gate.
A. AND B. OR C. NAND D. NOT
2. The _____ gate has two or more input signals. All inputs must be high to get a high output.
A. AND B. OR C. NAND D. NOT
3. _____ is way to simplify the equation.
A. Boolean Algebra B. K-MAP C. BOTH D. NONE
4. The OR gate has two or more input signals. If any input is _____, the output is high.
A. High B. Low C. Both A and B D. NONE
5. The relationship between a function and its binary variables can be represented in _____.
A. Truth table B. Decoder C. Encoder D. Multiplexer
6. A _____ is a combinational circuit that converts binary information from the 2^n coded inputs to a n outputs.
A. Half Adder B. Decoder C. Encoder D. Comparator
7. In Comparator, _____ gate is use for comparing bits in word.
A. XOR B. AND C. NOR D. XNOR
8. A combinational circuit that performs the arithmetic addition of two bits is called _____.
A. Full Adder B. Half Adder C. Binary Adder D. Decoder
9. Half adder is logic CKT that adds. _____ Digit at a time
A. 1 B. 2 C. 3 D. 4
10. In D flip-flop, when CLK is high then output is _____.
A. high B. low C. invert of input D. same as input

(P.T.O.)

Q2. Do as directed. (Attempt Any Ten)

[20]

1. Describe the NOT gate.
2. Draw the circuit for $A'B+B'C+AC'$
3. Write truth table for: $A'B+B'C$
4. Describe pair in k-map.
5. Simplify using k-map $F(A,B,C)=\Sigma(1,2,5)$
6. Explain commutative law.
7. Draw the circuit for comparator.
8. Draw the circuit for encoder
9. Define register.
10. Describe octet in k-map
11. Explain multiplexer in short.
12. Draw timing diagram of ring counter.

Q3. (A) Explain AND, OR gate.

[05]

(B) Explain de-morgan's first and second theorem with example.

[05]

OR

Q3. (A) Explain NOR, XNOR gate

[05]

(B) Explain associative and distributive law.

[05]

Q4. (A) Explain 8x3 line encoder in detail.

[05]

(B) Explain POS in detail.

[05]

OR

Q4. (A) Explain 3x8 line decoder in detail.

[05]

(B) Explain SOP in detail.

[05]

Q5. (A) Explain half adder in detail

[05]

(B) Explain binary subtractor in detail.

[05]

OR

Q5. (A) Explain full adder in detail.

[05]

(B) Explain binary adder in detail.

[05]

Q6. Explain D Flip-Flop in detail.

[10]

OR

Q6. Explain shift left and shift right register in detail.

[10]

—————X—————